KafkaGrpcFlow

A tiny social app to learn **how gRPC and Kafka fit together** in a microservice system.

Next.js → NestJS API Gateway → (gRPC) → Auth / Profile / Interaction services, each with its own Postgres. Events via Kafka using the Transactional Outbox pattern.

Run

Requires Node 20+ and Docker Desktop (running).

```
npm install
npm run infra:up    # Kafka :9092, Kafka UI :8080, Postgres :5433/:5434/:5435
npm run dev         # auth :50051, profile :50052, interaction :50053, gateway :3001, web
:3000
```

Open http://localhost:3000 · Kafka UI http://localhost:8080 · Stop infra and wipe data with `npm run infra:down`.

Docs (start here)

1. docs/overall.md: the whole flow, told through Riya's signup
2. docs/proto.md: the gRPC contracts
3. docs/gateway.md
4. docs/auth-service.md
5. docs/profile-service.md
6. docs/interaction-service.md
7. docs/grpc-vs-kafka.md: when to use which + interview Q&A
8. docs/idempotency-and-backpressure.md: duplicates, idempotency keys, rate limiting, overload